# Temporal Risk Identification and Behavioural Insight Mining in Mental Health: A Scalable Spark-SQL Based Predictive Pipeline

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## Abstract
This project focuses on investigating whether temporal and behavioural indicators can be used to gain insights, understand and predict the risk of employees having mental health issues using scalable Big Data Analytics. By developing a Spark-based data pipeline, we can access and study a large-scale mental health survey dataset from the Azure Blob Storage. The main purpose of this application is to identify and highlight the key time periods when the intervention is necessary for minimizing mental health problems. The Data extracted is processed in Apache Spark and the results/processed & cleaned data is stored in Postgres SQL database for evaluation and insights. Some of the challenges that were faced, revolved around noisy data, missing temporal fields and the integration of cloud were handled using schema validation, imputation strategies, and optimized Spark connectors. This study showcases how effectively these distributed systems can be configured and employed to help and provide assistance for early intervention and mental-health well-being in the corporate environments where the work pressure varies for different jobs.

**Keywords**: Temporal Indicators, Big Data Analytics, Apache Spark, Early Intervention, PostgreSQL

## I. Introduction
Despite growing awareness about the mental health in workplaces and organizations, these very companies lack and struggles to identify any sort of mental health problem, pressures and negative experiences proactively caused by the environment, work pressure, etc. Due to this, the graph for the mental health problems in employees are increasing with a steady rate. Existing methods in organizations such as surveys or one-off HR's check-ins are not good enough and cannot understand the depth & factors causing this mental stress increasing day by day amongst employees.

While these are just the overview of the problems some of the key problems that were identified were lack of temporal awareness and scalability of the existing approaches. Existing systems are weak at considering changes occurring over time, such as Gaps in mental health support and behavioural changes which are related to the work seasons or workload cycles. Also, these large-scale employee datasets are too big for traditional tools like Excel, or simple SQL which leads to not being processed, visualized or filtered for insights and is not integrated with real-time cloud environments. These factors result in having a dead system, which lacks any kind of warnings and is always dealt with reactively. Not, only that insights from these surveys of the employees remain underutilized for analysis which in turn increases the mental load on the employees.

To Deal with these problems, this study focuses on the whether temporal (e.g. time since last treatment or age) and behavioural indicators (e.g. treatment history, employer support) used to develop a model which can help and predict the employees whose health mental health is at risk. This is crucial as it helps in shaping new HR policies and providing mental health care which is proactive rather than the existing reactive systems. The novelty of this research analysis revolves around its prioritization of the temporal risks and behavior analysis of the employees which is designed using cloud integrated Spark pipeline, which offers a near real time infrastructure for predicting and analyzing mental health of the employees in organizations.

"Can temporal and behavioral indicators be used to develop a predictive model for identifying employees at high risk of mental health struggles, and what are the most vulnerable time periods during which interventions should be prioritized?"

The key characteristics of this problem from which this research is derived from are:

1. What are the key behavioral and demographic features that contribute significantly to the prediction of the mental health treatment in the workplace environment?
2. Are there specific hours or times of the day when then indicators with high mental health risk are at peak across individuals?
3. Can real-time prediction of the risks, combined with various temporal patters help organizations introduce proactive intervention methods more effectively?
4. How accurate is the predictive model in identifying individuals with high-risk mental health and how does its performance vary across different times. (e.g. weekday vs. weekends etc.)?

These questions not only provide a detailed understanding of the problem but also cover all aspects of it including the Angle of approach, Insights, Utility and Effectiveness of the solution. The answer to these questions paints a clear picture as to what is already implemented and what is still lacking helping our cause. It pushes us to focus on addressing these challenges with a Spark-powered, cloud-integrated, and automated pipeline, aiming to detect mental health risk patterns before critical stages, uncover when and why interventions should be prioritized, and build a foundation for predictive models using behavioral and temporal data for these organizations to proactively deal with mental health issues that the employees face.

## II. Related Work

The rapid rise of digital health technologies is redesigning the existing architecture of mental health management in workplace settings. The core of this transformation is the temporal and behavioral indicators which aim to identify individual/employees who are at a higher risk of mental distress using predictive models. Going through multiple recent research papers, a shared framework of integrating machine learning (ML), wearable technologies, and big data to support early intervention strategies has taken form.

Garriga et al. (2022) presents a machine learning model that is trained on electronic health records (EHRs) that predicts mental health crises accurately up to 28 days in advance. This study serves as a ground for the potential of a solution that uses temporal behavioral data for forecasting drastic psychological events. Complementing this, Meehan et al. (2022) systematically reviews over 300 psychiatric prediction models, noting the challenges that were faced like generalization and validation but designing models especially in complex real-world situations like an organization.

Koutsouleris et al. (2022) pushes the importance of ethical and interpretable AI in mental health applications. They support the systems that not only stick to the technical accuracy but go beyond that to include transparency, trust, and clinical relevance elements which are important for workplace adoption. The idea/concern of advocating system perspective aligns closely with that of Olawade et al. (2024), who emphasizes the need for AI-driven mental health systems, which are personalized and strictly abide by user trust and ethical integrity.

The role of infrastructure and automation in supporting predictive mental health systems is further developed by Muniswamaiah et al. (2019) and Gupta et al. (2017). Muniswamaiah et al. explores the integration of big data with cloud computing which outlines how cloud platforms enable the scalable storage and processing of huge behavioral and health datasets. This capacity enables storing sustainable real-time mental health data across different work environments. Gupta et al. introduces a big data framework which combines Apache Spark with deep learning to process and interpret the time series data efficiently. Their approaches focus on how distributed computing and automated model training pipelines can identify temporal and behavioral patterns which show the sings of psychological distress. With these studies combined, it provides a technical ground for implementing data-driven, automated mental health monitoring systems that are scalable, responsive and which are capable of continuous improvement with time.

Transformation of the findings of the research into scalability of the work environment requires a unique integration of big data frameworks with cloud infrastructure systems. Suspected behavioral indicators like stress and sleep patterns and heart rate are tracked through IoT and wearable devices according to Vimal et al. (2023) and Sabry et al. (2022). The data streams collect in centralized cloud-based platforms to monitor workplace populations persistently despite their distribution.

To process massive volumes of high-speed data one requires processing systems with strong capabilities. The real-time processing of streaming data demonstrates the usefulness of Apache Spark as described by Ebada et al. (2022) and Shrotriya et al. (2023). The distributed structure of Spark enables multiple processors to work simultaneously which makes it a perfect choice for analysis of extensive datasets which are done in real time. The built infrastructure working with PostgreSQL database can store historical trends which automatically triggers predictive model retraining. Organizations can create mental health intervention alerts using the data-driven dashboards by implementing automated ETL pipelines and data warehousing.

The automated infrastructure serves purposes of efficiency and forms the basis for developing workplace mental healthcare solutions that respond to dynamic data analysis. Behavioural and temporal insights become operational at scale through the combination of big data tools with cloud storage and scalable analytics to get predictive outputs from sensor-level data. Predictive mental health models achieve efficacy through Temporal and behavioural indicators using scalable ethical and interpretable systems. The available literature demonstrates how timely mental health services will integrate automation through cloud technologies to serve present-day workplace schedules.

## III. Methodology

This section highlights the detailed structure of the approach that we have used to deal with mental health big data in an automated pipeline setup using Spark. The goal is to make the architecture flexible, reproducible and easy to follow for scalability.

### A. Dataset Selection

The Dataset used is a mental health survey dataset, which is publicly available on Kaggle containing 200,000+ anonymized records. It contains indicators and features such as Age, Gender, Treatment Status, Employer Support, Support index, Risk Score etc. which provides information about the behavioral insights of the employee of a workplace.

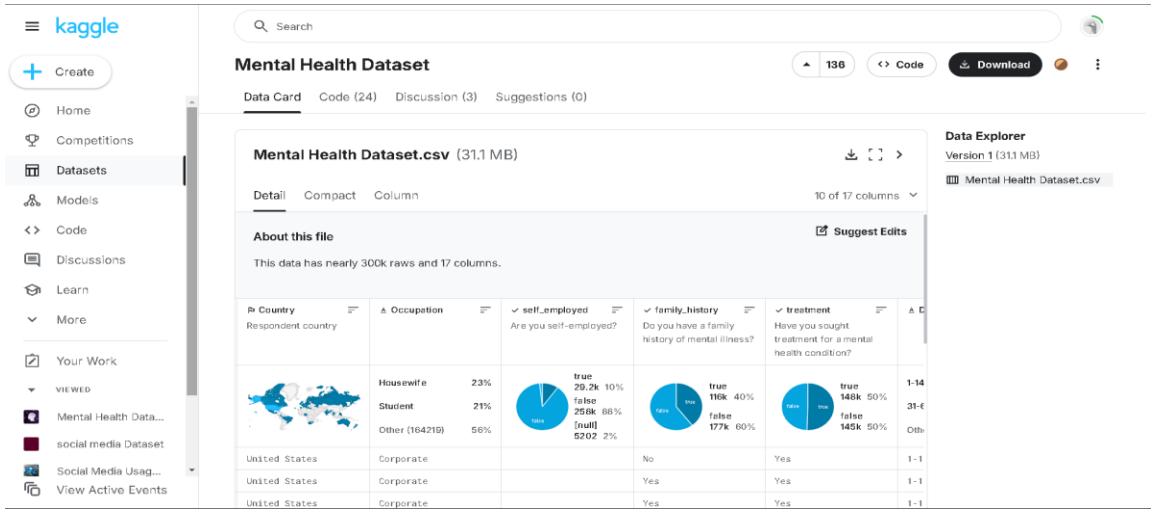


**Fig 1**. Kaggle Dataset for Mental Health

### B. Environment Setup

**i. Apache Spark (configured via PySpark):** Apache spark is an open-source distributed computing engine which is designed specifically for big data processing which allows memory computation across various clusters which increases the speed of processing significantly compared to the traditional MapReduce. PySpark is being used to process Big Data for mental health by loading and transforming data, reprocessing it and performing computations. It extracts the data from the Azure storage performing EDA and prepares it for structured storage.

**ii. PostgresSQL (via JDBC):** Postgres is an open-source relational database that supports SQL querying indexing and transactions. After cleaning the structured data is saved into Postgres SQL which allows us to query risk indicators and have easy integration with analytics tools like pandas and matplotlib.

**iii. Google Colab and Azure Blob Storage:** Google Colab is a cloud-based Python notebook, which gives us lightweight and easily accessible platform allowing us to run spark jobs. Microsoft Azure is being used which is a scalable cloud storage solution for unstructured data like CSV, logs and raw data sets. It stores raw data that can be easily retrieved into spark for processing.

### C. Data Processing

The Data is being processed using streamline automation pipeline using Spark with a cloud-based implementation of all the different steps.

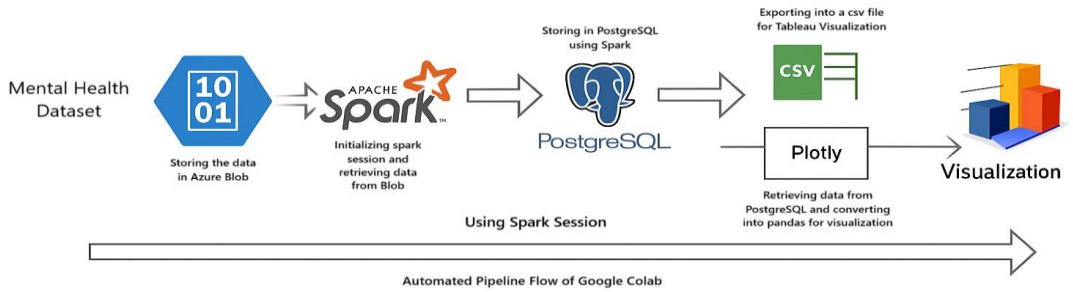


**Fig. 2** Automated Pipeline for processing Big Data

**i. Loading Data on Blob (Azure Cloud)**

The raw mental health dataset survey dataset is stored in Azure Blob Storage, which acts as a cloud-based data lake. It is ideal for storing big data since it is cost effective and offers scalability. It allows remote, centralized, and secure access to the raw dataset which is used by distributed processing engines like Spark as the next step for retrieving/ingesting data.

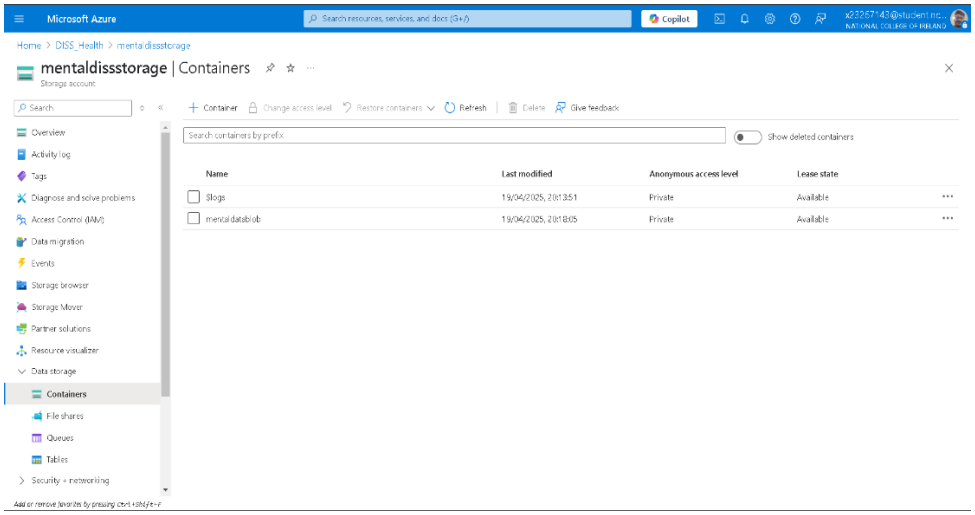


**Fig. 3** Storing raw data in Azure Blob

### ii. Retrieving Data using Spark

In this step, Spark creates a session which connects to the Azure Blob using Spark's Hadoop-compatible file system configuration (Fig. 4) using Apache Spark configured via PySpark in a Google Colab notebook. The dataset is then loaded from the Azure Blob storage into the Spark Dataframe which allows to handle big data using distributed data ingestion efficiently.

```
[ ] from pyspark.sql import SparkSession

    spark = SparkSession.builder \
        .appName("AzureToPostgres") \
        .config("spark.jars", "/content/postgresql-42.2.20.jar") \
        .config("spark.jars.packages",
                "org.apache.hadoop:hadoop-azure:3.3.1,"
                "org.eclipse.jetty:jetty-util-ajax:9.3.25.v20180904") \
        .getOrCreate()
```

```
account_name = "mentaldissstorage"
account_key = "BJXWDkvydwGmbAvY43IQHuHGf2mKDei3aulLDnNX8Js3IW6FtidoAiclFsCJ9fgBWuwZNR6HsweU+AStzK72oA=="
container_name = "mentaldatablob"
blob_name = "Mental Health Dataset.csv"

spark.conf.set(
    f"fs.azure.account.key.{account_name}.blob.core.windows.net",
    account_key
)

file_path = f"wasbs://{container_name}@{account_name}.blob.core.windows.net/{blob_name}"
```

**Fig. 4** Spark Session initialization and connection with Azure Blob

### iii. Data Preprocessing

This step includes all the cleaning/preprocessing operations which are required to prepare the dataset for further analysis helping us generate accurate insights. These have been done by using Spark distributed programming for the preprocessing of the large dataset which includes handling unknown, null & missing values, Normalizing string columns into Numeric values, Encoding categorical variables (Yes/No for treatment, support etc.), Engineering new features and schema validation & Summary of the datasets which provides valuable EDA and data quality and enhances the reliability of downstream predictions and visual insights.

### iv. Storing Data in PostgreSQL

Once the Data is processed and structured, it is then stored in PostgreSQL database using a JDBC connected from Spark. It is inserted into a defined Schema, allowing for persistent, structured and queryable storage, which helps in efficient data storage & retrieval, secure backup and compatibility with tools like Tableau and Power BI. The Fig. 5 shows the Loading of the cleaned data into PostgreSQL using JDBC.

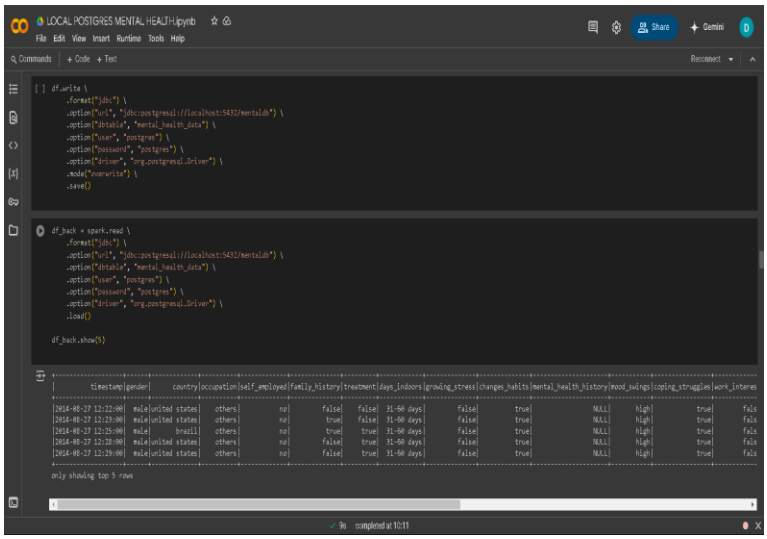


**Fig. 5** Loading and retrieving data from PostgresSQL using Spark

**v. Retrieving Data back from PostgreSQL**

In this step, cleaned and structured data is being retrieved from PostgreSQL for further analysis, which is done by converting the SQL table into Python's Pandas Dataframe for visualization. It allows us to move from distributed processing to lightweight visual analysis without running the Spark pipeline again.

**vi. Generating Visualizations**

Finally, the structured code is used to visualize insights which help in answering the core question of the research as to when and why intervention is necessary. Visualizing insights bridges the gap between data processing and actionable insights, aiding in identifying patterns and making decisions based on it. Seaborn, Matplotlib and Plotly are the libraries that have been used to visualize insights like Risk scores by age group, Support index heatmaps, Correlations between employer support, treatment, and temporal gaps etc.

**vii. Automation**

In order to make the pipeline hands-free and reusable, the execution of the process is done via notebook orchestration. Automation ensures scalability, reproducibility, and time efficiency which is essential in big data environments which also aligns with practices for continuous integration and deployment (CI/CD). When the run all is clicked, all the above-mentioned processes run in a single pipeline resulting in a automated flow.

## D. Structure of Solution

The solution is defined as a scalable and automated pipeline which follows ETL (Extract, Load and Transform) pattern, with the integration of cloud storage and distributed processing. The flow shows the feasibility and the ease of use of the cloud implemented solution which can be accessed and reproduced for further use. The data travels through various stages from being raw to structured for analysis, which enables the use of big data without worrying about the local storage and the system setup. It is not only cost efficient but can be implemented and automated in real time for the use case. Thus, this design architecture of the flow is suitable for being used as a solution for obtaining mental health insights from the employees in real time.

## IV. Results and Evaluation

This section helps in understanding the insights provided by the clean data to understand the temporal and behavior indicators through visualizing various aspects of the data. These insights directly inform our research question regarding risk prediction and intervention timing.

**i. Heatmap of mental health treatment by Hour and Weekday**

The heatmap shows the distribution of treatment-seeking behavior across different hours of the day and days of the week. From the graph we can tell that the Treatment rates are consistently higher during late evenings and early mornings, particularly on weekends and minimal activity in early weekdays between 9 AM and 12 PM which indicates low intervention response. These insights suggest that the intervention programs should be scheduled outside of typical work hours, like during the evening times or weekends to make it efficient and engaging for the employees.

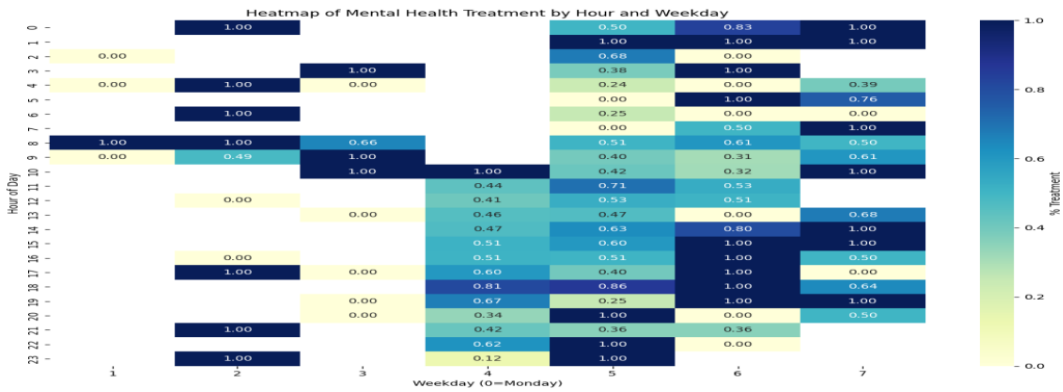


**Fig. 6** Heatmap of mental health treatment by Hour and Weekday

**ii. Daily Trends of Mental Health Stress Indicators**

The below line graph shows temporal indicators such as growing stress, mood swings, and coping struggles and how it changes over time. It is evident that Mood swings remain consistently high, while coping struggles fluctuates significantly, which might have been influenced by environmental or workplace triggers. Growing stress shows spikes at irregular intervals, indicating seasonal or workload-related stress peaks which confirms the idea of the monitoring temporal patterns to identify vulnerable periods where interventions could be more impactful.

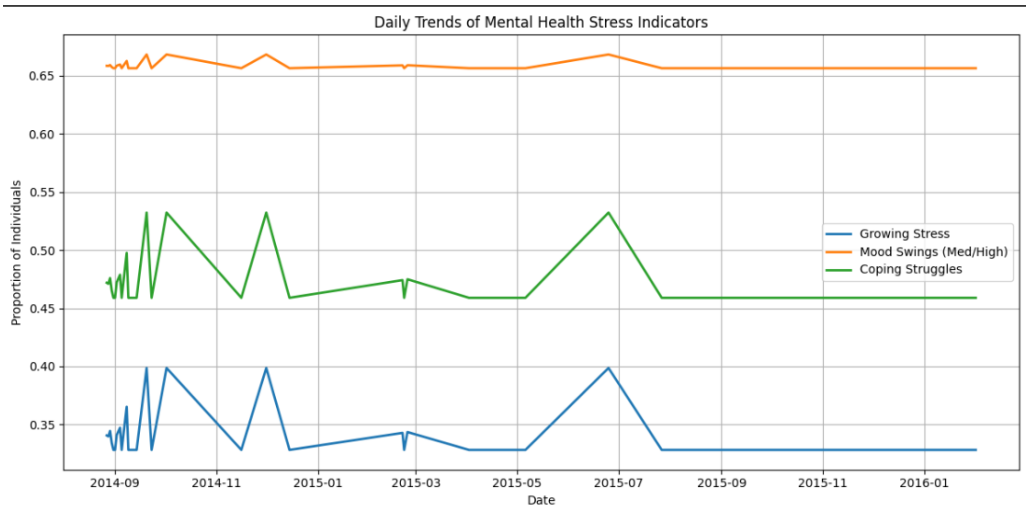


**Fig. 7** Daily Trends of Mental Health Stress Indicators

**iii. Feature Importance for Predicting Mental Health Treatment**

This bar chart provides an overview of how much each behavioral indicator contributes to predicting whether an individual needs treatment or not. It can be derived from the graph that Days spent indoors is the strongest predictor

which suggests that isolation of the employees is a major concern for their mental health with other indicators like growing stress, coping struggles, and social weakness also show less influence. These insights validate the use of behavioral indicators as features in predictive models and provide direction for feature selection in future ML pipelines.

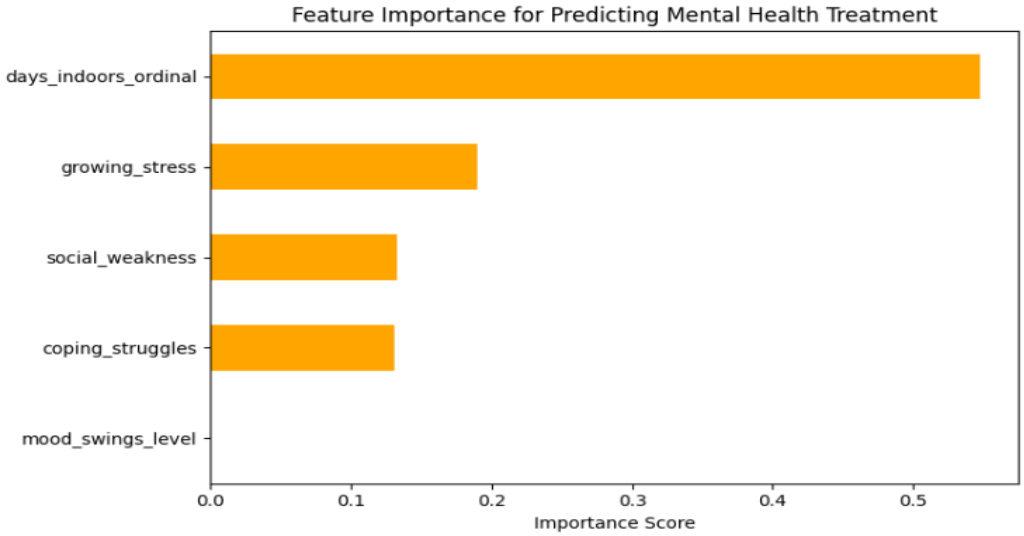


**Fig. 8** Feature Importance for Predicting Mental Health Treatment

**iv. STL Decomposition of Weekly Treatment Rate**

The STL decomposition graph shows all the components like seasonal, trend, and residual for treatment trends. The seasonal component shows weekly variations which means that the treatment seeking behavior is not random. Also, the read remains relatively stable indicating the need for mental health treatment remains consistent. These patterns in trends can help the organization understand and align mental health resources more effectively making it more proactive.

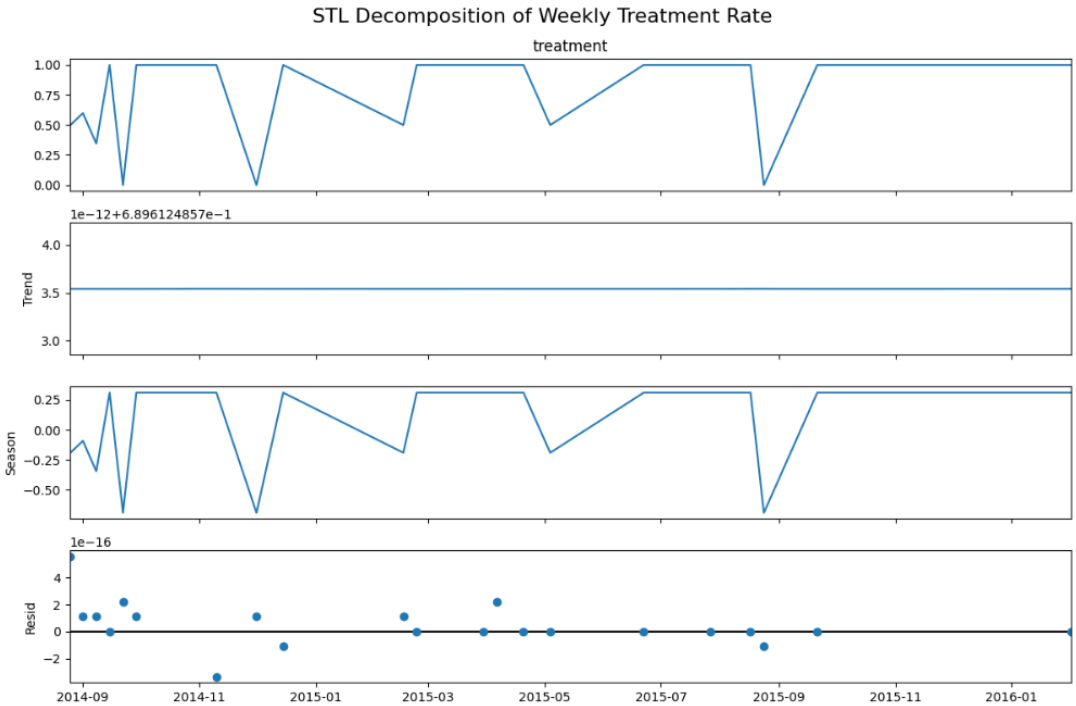


**Fig. 9** STL Decomposition of Weekly Treatment Rate

**v. Correlation Heatmap of Mental Health Risk Features**

The heatmap graph shows the relationship between features like stress, mood, coping and the actual treatment status. It is evident that the correlation values are modest but there are dependencies between the behavioural indicators. The sight correlation indicates towards a combination of weak relations across features, which calls for multi variate modelling approaches.

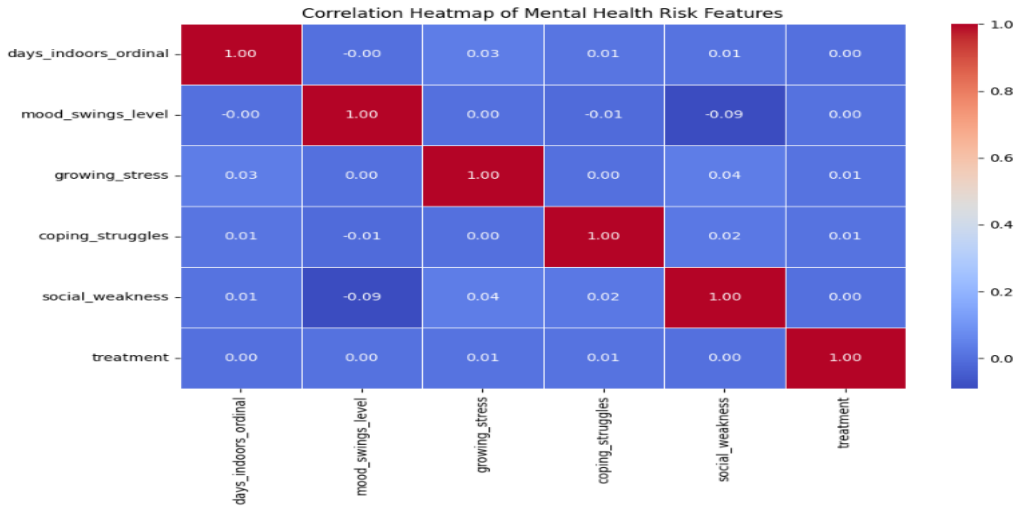


**Fig. 10** Correlation Heatmap of Mental Health Risk Features

## vi. Dashboard Visualization

The tableau dashboard provides information about the trends and the behavioral risks associated with the individual's mental health on a global scale. The weekly treatment trends highlight noticeable fluctuations which can be an indication of temporal vulnerability points. The country-level treatment map shows the geographic disparities, while the behavioural risk breakdown by treatment status reveals which psychological factors are most associated with individuals who are not treated. Together, these insights support the research question by showing how behavioural and temporal indicators can guide targeted mental health interventions.

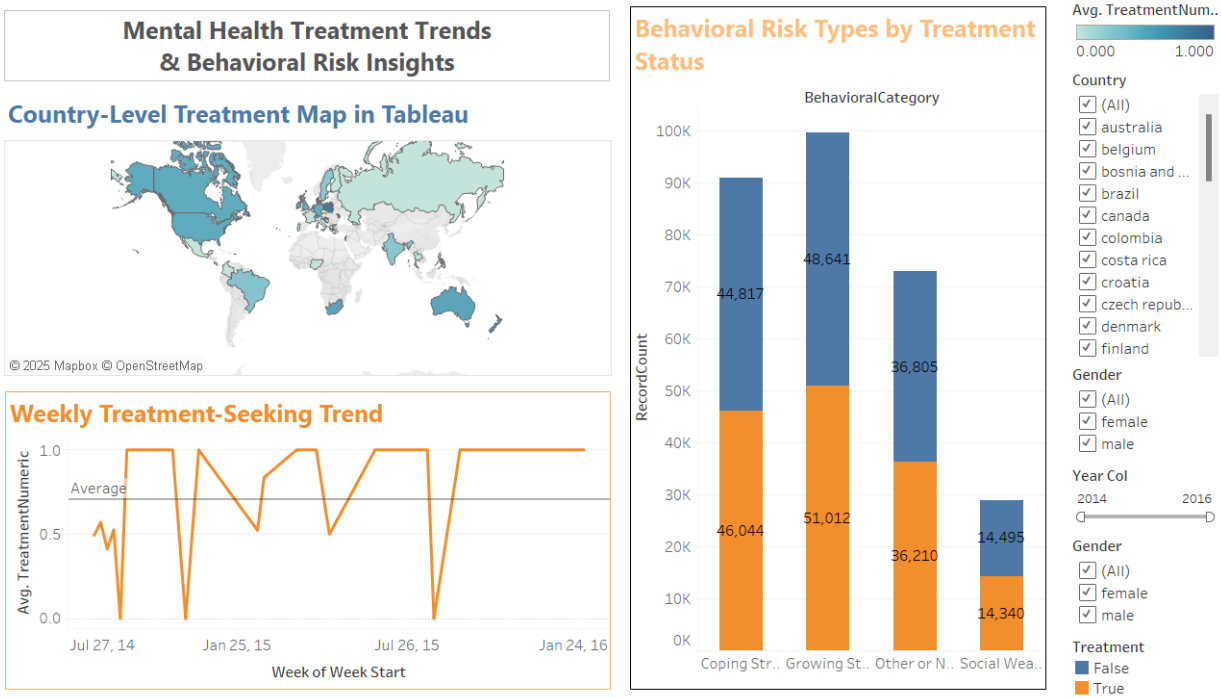


**Fig. 11** Mental Health Treatment Trends and Behavioral Risk Insights

## V. Conclusion and Future Work

This project successfully developed a cloud-based big data automated pipeline using Apache Spark and PostgreSQL to analyse temporal and behavioural indicators of employee mental health risk. Distributed processing through PySpark allowed us to store clean data into PostgreSQL enabling us to explore the large-scale survey data stored in Azure BLOB storage efficiently. The above visual analysis provided key insights like treatment seeking behavior fluctuates by time of the day and week, while features such as days spent indoors, growing stress, and coping struggles were the key behavioral predictors in understanding and predicting the mental health of the employees.

The results demonstrate how temporal, and behavior indicators are such valuable signals for identifying individuals with higher risk of mental health and the time periods where the interventions will be impactful. Based on these findings it is evident that the feasibility of building a predictive model to assist organizations in early proactive detection of the mental health support of the employees is supported.

Some of the future work slash enhancements of this project can include:

- Using and deploying a machine learning model using spark ML library or scikit learn to predict mental health risks based on the temporal and behavioral indicators which can assist in understanding and preventing it in future.
- On the visualization side, deployment of an interactive dashboard for HR professionals will offer risks alert, heat maps and intervention recommendations based on the featured parameters.
- Incorporation of NLP techniques to analyze survey responses and text reviews . This will help us to understand the emotional tone and context enabling us to detect distress

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